Category Theory for Mathematical Physics — Exercise Sheet 13

Boundaries, defects and non-invertible symmetries

Variant: Solutions

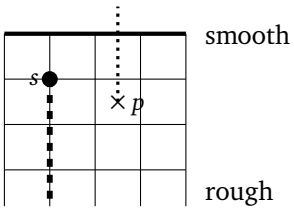
Conventions as in Lecture 13: the toric code has $A_s = \prod X$ on the edges at a vertex and $B_p = \prod Z$ around a face; $e = (0, \chi_-)$, $m = (1, 1)$, $\varepsilon = e \otimes m$ (here ε is the fermion), and its anyons are as in Theorem 12.8. An algebra is (L, μ, ι) and $d(L)$ its dimension. On the Ising chain of n sites, $\eta = \prod_j X_j$, and $\mathcal{T}$ is the translation, $\mathcal{T} O_j \mathcal{T}^{-1} = O_{j+1}$ (not the modular T); D is the Kramers–Wannier operator of § 13.3, and indices are mod n. $\zeta = e^{2\pi i/3}$, $\varphi = (1 + \sqrt{5})/2$, and ω is a 3-cocycle. In Exercise 6, $\varepsilon(b, c)$ is instead Sheet 3's carry, $\zeta_4 = i$, and ψ is a 2-cochain.

Exercise 1 (The toric-code boundaries on the lattice). (i) Draw a piece of the lattice with a smooth boundary on top and a rough one below. On it draw a string of Z's from a bulk vertex s down to the end of a dangling edge, and a string of X's on the edges crossed by a dual path from a bulk face p up through the smooth boundary. Show that each string violates only the term at its bulk end.

(ii) Show that a Z-string from s that ends at a vertex of the smooth boundary violates the star there too, and that an X-string from p that leaves through the rough boundary violates a three-edge plaquette. What does ε, brought to each boundary, leave behind?

(iii) The *dual lattice* has a vertex in each face and an edge across each edge. Show that passing to it, and exchanging X with Z, turns stars into plaquettes and back, the smooth boundary into a rough one, and e into m.

Solution. (i) In the picture the Z-string runs down the dashed vertical path from s, the X-string on the edges crossed by the dotted dual path from p:



A product of Z's on a set b of edges anticommutes with A_v exactly when an odd number of edges of b end at v. Along the path every vertex other than s meets two of its edges, and the lower end of the dangling edge is no vertex (it has no star), so only A_s is violated: the bulk e at s remains, and the end on the rough boundary leaves no excitation. Likewise a product of X's anticommutes with B_f exactly when an odd number of its edges lie on ∂f; along the dual path every face other than p contains two crossed edges, and the top boundary edge lies in one face only, so only B_p is violated: m is absorbed at the smooth boundary.

(ii) A Z-string ending at a vertex v of the top row meets v in one edge, so A_v (a three-edge star) is violated: e arrives at the boundary as an excitation. An X-string leaving through the bottom crosses the horizontal edge of a three-edge plaquette, which it meets once, so that plaquette is violated. So $\varepsilon = e \otimes m$ brought to the smooth boundary leaves e, and to the rough boundary m: the maps $1, m \mapsto 1, e, \varepsilon \mapsto e$ and $1, e \mapsto 1, m, \varepsilon \mapsto m$ of Lecture 13.

(iii) Dual vertices are faces, dual edges cross edges, and a dual face surrounds a vertex; the edges around a dual face are those crossing the edges at that vertex. So, with $X \leftrightarrow Z$,

the star $A_v = \prod X$ on the edges at v becomes $\prod Z$ around the dual face at v, a plaquette, and B_f becomes a star at the dual vertex f: the Hamiltonian is the toric code again. A top edge lies in one face, so its dual edge has one end, at that face, and dangles: the smooth boundary becomes rough, and its three-edge stars become three-edge plaquettes. A violated star, e, becomes a violated plaquette, m. This is the lattice face of the exchange $e \leftrightarrow m$ of Sheet 12, Exercise 4: it carries one boundary to the other.

Exercise 2 (Lagrangian algebras). (i) The double semion $\mathcal{Z}(\mathrm{Vec}^{\omega}_{\mathbb{Z}_2})$, $\omega(1,1,1) = -1$, has the four simple objects $(\delta_0, \pm 1)$, $(\delta_1, \pm i)$ (Sheet 12, Exercise 6; spins as in Lecture 12), all invertible of dimension 1, with spins $1, 1, i, -i$ (the self-braidings). Let $b = (\delta_0, -1)$. Show that $b \otimes b \cong 1$, and, as in the proof of Theorem 13.3, that the only Lagrangian algebra is $1 \oplus b$. How does this match Theorem 13.4 and the module categories over $\mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$ (Lecture 11)?

(ii) In $\mathcal{Z}(\mathrm{Fib}) \simeq \mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ the simple objects 1, $\tau \boxtimes 1$, $1 \boxtimes \tau$, $\tau \boxtimes \tau$ have $d = 1, \varphi, \varphi, \varphi^2$ (Sheet 12, Exercise 5). Show that a connected L with $d(L)^2 = \dim \mathcal{Z}(\mathrm{Fib})$ is $1 \oplus (\tau \boxtimes \tau)$ as an object. With Sheet 11, Exercise 4(iii), conclude that $\mathcal{Z}(\mathrm{Fib})$ has exactly one Lagrangian algebra, and, by Theorem 13.5 and the physical principle of Lecture 13, exactly one gapped boundary.

Solution. (i) b lies in degree 0, so every associator in the half-braiding of $b \otimes b$ is a value of ω with an argument 0, which is 1; the half-braiding of $b \otimes b$ on δ_h is then $(-1)^h(-1)^h = 1$, and $b \otimes b = (\delta_0, 1) = 1$. Now let L be Lagrangian: $\dim \mathcal{Z} = 4$, so $d(L) = 2$, and connectedness leaves $L = 1 \oplus X$ with $X \neq 1$ invertible. If $X \otimes X \ncong 1$ (which in fact does not happen here: all four objects square to 1), then $X \otimes X$ is a simple object other than 1 and X (as $X \neq 1$), so $\mathrm{Hom}(X \otimes X, L) = 0$: the product vanishes on $X \otimes X$, and, as for $\lambda = 0$ in the proof of Theorem 13.3, X is a submodule of L with no complement, so L is not separable. So $X \otimes X \cong 1$, and the product there is $\lambda \neq 0$, rescaled to 1; on $X \otimes X \cong 1$ the braiding $c_{X,X}$ acts by the self-braiding θ_X, so commutativity needs $\theta_X = 1$. Of b and the two semions only b has $\theta = 1$, and for $X = b$ the proof of Theorem 13.3 applies word for word (the section Δ, commutativity). So $1 \oplus b$ is the only Lagrangian algebra. By Theorem 13.4 it matches one indecomposable module category over $\mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$: there is only the regular one, since $\mathrm{Vect}_{\mathrm{fd}}$ with δ_1 acting needs $\omega(1,1,1) = 1$ (Theorem 11.6 and the remark after it). By Theorem 13.5 ($\mathrm{Vec}^{\omega}_{\mathbb{Z}_2}$ is unitary) and the physical principle, the double semion has exactly one gapped boundary, at which b condenses.

(ii) $\dim \mathcal{Z}(\mathrm{Fib}) = (1+\varphi^2)^2$, so $d(L) = 1 + \varphi^2$. Connected means 1 appears once: $L = 1 \oplus a(\tau \boxtimes 1) \oplus b(1 \boxtimes \tau) \oplus c(\tau \boxtimes \tau)$ with $a, b, c \geq 0$, and $d(L) = 1 + (a+b)\varphi + c\varphi^2 = 1 + (a+b+c)\varphi + c$. Equating with $1 + \varphi^2 = 2 + \varphi$ gives $(a+b+c-1)\varphi = 1 - c$; as φ is irrational, $c = 1$ and $a + b = 0$. So $L = 1 \oplus (\tau \boxtimes \tau)$, whose summand has spin $\theta_\tau \bar{\theta}_\tau = 1$. By Sheet 11, Exercise 4(iii), Fibonacci has exactly one indecomposable module category, the regular one, so by Theorem 13.4 $\mathcal{Z}(\mathrm{Fib})$ has exactly one Lagrangian algebra, and it is this object. By the physical principle, with Theorem 13.5 for its existence (Fibonacci is unitary), $\mathcal{Z}(\mathrm{Fib})$ has exactly one gapped boundary, at which $\tau \boxtimes \tau$ condenses; $\tau \boxtimes 1$ and $1 \boxtimes \tau$ do not.

Exercise 3 (Condensing e). Let $L = 1 \oplus e$ in $\mathcal{D} = \mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$, with the product of Theorem 13.3. A right L-module (M, ρ) is *local* if $\rho \circ c_{L,M} \circ c_{M,L} = \rho$. You may use: every L-module is a direct summand of a free one, $X \otimes L$ with $\rho = \mathrm{id}_X \otimes \mu$ (L is separable); the local modules form a braided fusion category $\mathcal{D}^{\mathrm{loc}}_L$, the phase left after condensing L, with unit L; and, as L is commutative, $X \otimes L$ with X simple is local exactly when the double braiding $c_{L,X} \circ c_{X,L}$ is the identity of $X \otimes L$.

(i) Show that $e \otimes L \cong L$ and $\varepsilon \otimes L \cong m \otimes L$ as L-modules, and that L and $m \otimes L$ are simple modules.

(ii) Show that L is local and $m \otimes L$ is not.

(iii) Conclude that $\mathcal{D}_L^{\mathrm{loc}} \simeq \mathrm{Vect}_{\mathrm{fd}}$: condensing e leaves the trivial phase. What happens to m and ε?

Solution. (i) The map $f = \mu \circ (\iota_e \otimes \mathrm{id}_L) \colon e \otimes L \to L$, with $\iota_e \colon e \to L$ the inclusion, is a module map by associativity of μ. On components it is $e \otimes 1 \to e$ and $e \otimes e \to 1$, both isomorphisms ($\lambda = 1$), so f is an isomorphism: $e \otimes L \cong L$. Tensoring with m on the left, $\varepsilon \otimes L = m \otimes e \otimes L \cong m \otimes L$. A nonzero proper submodule of L would be 1 or e; but 1 acted on by e gives e, and e acted on by e gives 1, so neither is closed: L is simple. In the same way $m \otimes L = m \oplus \varepsilon$, and the action of e exchanges the two summands, so $m \otimes L$ is simple. As every module is a summand of some $X \otimes L$, the simple modules are L and $m \otimes L$: two, as Theorem 13.4 says ($\{1, m\}$, the rough boundary's excitations).

(ii) $c_{L,X} \circ c_{X,L}$ acts on the summand $X \otimes a$, $a \in \{1, e\}$, by the double braiding of X and a. For $X = 1$ all are 1: L is local. For $X = m$ the double braiding with e is -1 (Sheet 12, Exercise 1), so $m \otimes L$ is not local.

(iii) $\mathcal{D}_L^{\mathrm{loc}}$ has one simple object, its unit L, with $\mathrm{End}(L) = \mathrm{Hom}_L(L, L) \cong \mathrm{Hom}(1, L) = \mathbb{C}$ (L connected): it is $\mathrm{Vect}_{\mathrm{fd}}$, the trivial phase, with no anyons. Condensing e makes e part of the vacuum; m and ε, which braid with -1 around e, cannot exist alone in the condensate and are confined (they survive only at the end of a line, as the boundary excitation m). The dimensions agree: $\dim \mathcal{D}/d(L)^2 = 4/4 = 1$. The rough boundary, where e condenses, is thus a wall between the toric code and the vacuum.

Exercise 4 (The four boundaries of $\mathcal{Z}(\mathrm{Vec}_{S_3})$). Write t for the class of transpositions and u for that of 3-cycles, and the eight anyons of Sheet 12, Exercise 3, as $1_{\mathbf{1}}, 1_{\mathrm{sgn}}, 1_{\mathrm{std}}$ (class $\{1\}$), $t_\pm$ and $u_1, u_\zeta, u_{\zeta^2}$ (the character of the centraliser at the class), with dimensions $1, 1, 2, 3, 3, 2, 2, 2$ and spins $1, 1, 1, 1, -1, 1, \zeta, \zeta^2$.

(i) Explain why $\mathcal{Z}(\mathrm{Vec}_{S_3})$ has exactly four Lagrangian algebras, and, by the physical principle, four gapped boundaries.

(ii) You may use (Beigi, Shor and Whalen, arXiv:1006.5479, §4.1, eq. (19)) that the boundary built from $\mathrm{Mod}(\mathbb{C}[H])$, $H \leq S_3$, condenses the anyons of the $D(S_3)$-module A_H with basis the pairs (xH, g), $g \in xHx^{-1}$, of flux g, on which $k \in S_3$ acts by $k \cdot (xH, g) = (kxH, kgk^{-1})$; the multiplicity of $([g], \pi)$ in A_H is that of π in the permutation representation of the centraliser of g on the pairs of flux g. Decompose A_H for the four classes of H.

(iii) Check that each A_H has dimension $6 = \sqrt{\dim \mathcal{Z}(\mathrm{Vec}_{S_3})}$ and only summands of spin 1. Which boundary condenses only pure charges, and which only pure fluxes?

Solution. (i) Over Vec_{S_3} there are four indecomposable module categories, $\mathrm{Mod}(\mathbb{C}[H])$ for the four classes of subgroups $1, \mathbb{Z}_2, \mathbb{Z}_3, S_3$ (Sheet 11, Exercise 3). By Theorem 13.4 they match four Lagrangian algebras in $\mathcal{Z}(\mathrm{Vec}_{S_3})$, and by Theorem 13.5 and the physical principle these are the four gapped boundaries.

(ii) The pairs of flux g are the cosets xH with $x^{-1}gx \in H$, and the centraliser $C(g)$ permutes them by left multiplication. $H = 1$: only $g = 1$, with all six cosets; S_3 acts by its regular representation, $\mathbf{1} \oplus \mathrm{sgn} \oplus 2\,\mathrm{std}$, so $A_1 = 1_{\mathbf{1}} + 1_{\mathrm{sgn}} + 2 \cdot 1_{\mathrm{std}}$. $H = \mathbb{Z}_2 = \langle (12) \rangle$: for $g = 1$ the three cosets, permuted by S_3 as $\mathbf{1} \oplus \mathrm{std}$; for $g = (12)$ only the coset H ($x^{-1}(12)x = (12)$ means $x \in H$), fixed by $C(g) = H$: the trivial character; 3-cycles never. So $A_{\mathbb{Z}_2} = 1_{\mathbf{1}} + 1_{\mathrm{std}} + t_+$. $H = \mathbb{Z}_3$: for $g = 1$ the two cosets, permuted as $\mathbf{1} \oplus \mathrm{sgn}$; for $g = (123)$ both cosets (as $\mathbb{Z}_3$ is normal), and $C(g) = \mathbb{Z}_3$ fixes them both: twice the trivial character; transpositions never. So $A_{\mathbb{Z}_3} = 1_{\mathbf{1}} + 1_{\mathrm{sgn}} + 2u_1$. $H = S_3$: one coset, and every g; each centraliser fixes it, so $A_{S_3} = 1_{\mathbf{1}} + t_+ + u_1$.

(iii) The dimensions are $1+1+2\cdot 2$, $1+2+3$, $1+1+2\cdot 2$ and $1+3+2$, all $6=\sqrt{36}$, as a Lagrangian algebra requires, and every summand has spin 1 (t_+ and u_1 carry the trivial character), consistent with commutativity (that every summand of a connected étale algebra has spin 1 is cited; Theorem 13.3 proves it for invertible summands). The boundary of $H=1$, the regular module, condenses only the pure charges 1_π, each $\dim\pi$ times: the analogue of the rough boundary. That of $H=S_3$, the fibre functor $\mathrm{Vect}_{\mathrm{fd}}$, condenses one pure flux of each class, with trivial charge: the analogue of the smooth one. The other two mix charges and fluxes.

Exercise 5 (Kramers–Wannier on the ring). Use the relations $DX_j = Z_{j-1}Z_jD$, $DZ_{j-1}Z_j = X_{j-1}D$ and $\eta D = D$ of § 13.3, on a ring of $n\geq 3$ sites.

(i) Show that $D^2X_j = X_{j-1}D^2$ and $D^2Z_{j-1}Z_j = Z_{j-2}Z_{j-1}D^2$, and that $\mathcal{T}D^2$ commutes with every X_j and every $Z_{j-1}Z_j$.

(ii) Show that an operator commuting with every X_j and every $Z_{j-1}Z_j$ is $a+b\eta$ for numbers a,b.

(iii) Deduce that $D^2 = a(1+\eta)\mathcal{T}^{-1}$ (Seiberg, Seifnashri and Shao normalise D so that $a=1$). Why is this not the fusion rule $N\otimes N = 1\oplus\eta$ of a fusion category?

(iv) In *Integrability*, Lecture 18, the chain is $H=-J\sum_k\sigma^x_k\sigma^x_{k+1}-h\sum_k\sigma^z_k$, with parity $\Pi=\prod_k\sigma^z_k$ (its site written k here). Write the Kramers–Wannier relations and D^2 in those letters. Shao uses $D_{\mathrm{Shao}} = D^\dagger/\sqrt{2}$, with D normalised as in (iii), and his T_{Ising} moves O_j to O_{j+1}: compute D^2_{Shao}.

Solution. (i) $D^2X_j = D(Z_{j-1}Z_jD) = X_{j-1}D^2$ and $D^2Z_{j-1}Z_j = D(X_{j-1}D) = Z_{j-2}Z_{j-1}D^2$: D^2 moves these generators, and so every operator built from them (the η-even ones), one site back. It does not do so for a single Z_j, which D does not map to a local operator. With $\mathcal{T}O_{j-1}\mathcal{T}^{-1}=O_j$, $\mathcal{T}D^2X_j = \mathcal{T}X_{j-1}D^2 = X_j\mathcal{T}D^2$, and likewise for $Z_{j-1}Z_j$.

(ii) The X_j commute, so their joint eigenvectors $|x\rangle$, $x\in\{\pm1\}^n$, form a basis, each eigenspace a line. An operator P commuting with every X_j preserves each line: $P|x\rangle = f(x)|x\rangle$. The operator $Z_{j-1}Z_j$ anticommutes with X_{j-1} and X_j and commutes with the others, so it maps $|x\rangle$ to (a multiple of) $|x'\rangle$, with x_{j-1},x_j flipped; P commuting with it gives $f(x')=f(x)$. Flipping neighbouring pairs connects any two x with the same product $\prod_j x_j$ (move the flipped signs together along the ring), so $f(x)=a+b\prod_j x_j$, that is $P=a+b\eta$.

(iii) By (i) and (ii), $\mathcal{T}D^2 = a+b\eta$. Also $\eta D^2 = D^2$, and η commutes with $\mathcal{T}$, so $\eta(a+b\eta) = a+b\eta$, that is $a\eta+b = a+b\eta$, and $a=b$ (η is not a number). So $D^2 = a(1+\eta)\mathcal{T}^{-1}$. The fusion rule $N\otimes N = 1\oplus\eta$ would need $D^2 = 1+\eta$, a sum of lines of the symmetry; here it carries $\mathcal{T}^{-1}$, a translation, which is not an internal symmetry of the chain but moves the sites themselves. And $\mathcal{T}$ is not 1 on the η-even states: in the X-eigenbasis, $|+\cdots+\rangle$ is η-even and fixed by $\mathcal{T}$. If $\mathcal{T}$ were a scalar c on the even states, then $c=1$, and for the even state $v = Z_0Z_1|+\cdots+\rangle$, $v = \mathcal{T}v = Z_1Z_2\mathcal{T}|+\cdots+\rangle = Z_1Z_2|+\cdots+\rangle$; but for $n\geq 3$ these flip different pairs of signs. So $\mathcal{T}$ is not a scalar, in particular not 1, on the even states. (At $n=2$, $\mathcal{T}$ is the swap, which is 1 on the even states, and $D^2=1+\eta$ there by accident.) So the lattice operators do not realise the fusion rule; only in the continuum, where $\mathcal{T}\to 1$ on the relevant states, does $N\otimes N = 1\oplus\eta$ hold (Theorem 13.7).

(iv) With $X\leftrightarrow\sigma^z$, $Z\leftrightarrow\sigma^x$, $\eta=\Pi$ and $g=h/J$ ($H = J\,H_{\mathrm{chain}}$), and the site written k: $D\sigma^z_k = \sigma^x_{k-1}\sigma^x_kD$, $D\sigma^x_{k-1}\sigma^x_k = \sigma^z_{k-1}D$, $\Pi D = D$ and $D^2 = (1+\Pi)\mathcal{T}^{-1}$; it maps the field term to the bond term, $h\leftrightarrow J$, as in *Integrability*'s Prop. 18.9. For Shao's operator, $D^2_{\mathrm{Shao}} = \frac12(D^2)^\dagger = \frac12\big((1+\eta)\mathcal{T}^{-1}\big)^\dagger = \frac12\mathcal{T}(1+\eta) = \frac12(1+\eta)\mathcal{T}$, using $(\mathcal{T}^{-1})^\dagger = \mathcal{T}$ ($\mathcal{T}$ is unitary) and $\eta^\dagger=\eta$. So $T_{\mathrm{Ising}} = \mathcal{T}$, and the translation appears with the opposite power because D_{Shao} is built from

$D^\dagger$, not because the two translations differ. (The script builds D from Shao's formula on rings of 3 to 6 sites and checks the three relations used, (i), $\mathcal{T}D^2 = 1+\eta$ and D^2_{Shao}.)

Exercise 6 (Full gauging and the anomaly). (i) For the cocycles ω_p of $\mathbb{Z}_4$ (Sheet 3), $p \in \{0,1,2,3\}$, use the invariant $Q_\omega(1) = \prod_{r=0}^{3} \omega(1,r,1)$, which is unchanged when ω is multiplied by a coboundary, to show that $\mathrm{Vec}^{\omega_p}_{\mathbb{Z}_4}$ is anomalous for $p \neq 0$. Which p allow a rank-one module category over the part on $\{0,2\}$ (Sheet 11, Exercise 2)?

(ii) Gauging all of Vec_{S_3}: using $H^2(S_3, \mathbb{C}^\times) = 0$ (given, as in Sheet 11, Exercise 3), how many choices of ψ are there, and what are the simple lines of the dual symmetry? Show that std has no inverse.

Solution. (i) $\varepsilon(r,1) = 1$ only for $r = 3$, so $Q_{\omega_p}(1) = \omega_p(1,3,1) = \zeta_4^p = i^p$. A coboundary has $Q = Q_{\omega_0} = 1$, so for $p \neq 0$ the class $[\omega_p]$ is nonzero and, by Definition 13.8, $\mathrm{Vec}^{\omega_p}_{\mathbb{Z}_4}$ is anomalous: its $\mathbb{Z}_4$ symmetry cannot be gauged in full. The part on $\{0,2\}$ has a rank-one module exactly for p even (Sheet 11, Exercise 2): for $p = 2$ the anomaly does not reach the subgroup $\{0,2\}$, whose restricted cocycle is a coboundary.

(ii) By Theorem 13.9 the choices of ψ differ by classes in $H^2(S_3, \mathbb{C}^\times)$, which is 0 (given): one. The dual symmetry is $(\mathrm{Vec}_{S_3})^*_{\mathrm{Vect}_{\mathrm{fd}}} \simeq \mathrm{Rep}(S_3)$, with the simple lines $\mathbf{1}, \mathrm{sgn}, \mathrm{std}$. If $\mathrm{std} \otimes V \cong \mathbf{1}$, dimensions give $2 \dim V = 1$, impossible; indeed $\mathrm{std} \otimes \mathrm{std} \cong \mathbf{1} \oplus \mathrm{sgn} \oplus \mathrm{std}$ (Lecture 3). Gauging the ordinary symmetry S_3 produces a non-invertible one.

Exercise 7 ⋆. *Ground states on a cylinder.* Put the toric code on a cylinder: circumference $W \geq 2$, vertex rows $y = 0, \ldots, \ell$ ($\ell \geq 1$), and a smooth or rough boundary at each end (a rough end has dangling edges and three-edge plaquettes). You may use *Quantum Computing*'s counting: r independent commuting Pauli generators on n qubits, with $-\mathbb{1}$ not in the group they generate, fix a space of dimension 2^{n-r}.

(i) With a rough boundary below and a smooth one on top, count the qubits, stars and plaquettes, and show that the ground space is one-dimensional.

(ii) Do the same with both ends rough, and with both smooth: the ground space has dimension 2.

(iii) Compare with the number of anyons that condense on both boundaries, counted with multiplicity,

$$\dim \mathrm{Hom}(L_{\text{top}}, L_{\text{bottom}}),$$

for $L = 1 \oplus e$ (rough) and $1 \oplus m$ (smooth). Which string operators act on the two ground states when both ends are rough?

Solution. (i) The horizontal edges are $W(\ell+1)$, and the vertical ones $W\ell$ inside plus W dangling below: $n = 2W(\ell+1)$. There are $W(\ell+1)$ stars and $W\ell + W = W(\ell+1)$ plaquettes (the bottom ones with three edges): $2W(\ell+1) = n$ generators. They commute, as on the torus. A product of the stars at a set V of vertices is $\prod X$ over the edges with an odd number of ends in V; it is $\mathbb{1}$ only if no such edge exists. A dangling edge has one end, so no bottom vertex lies in V; an edge with both ends at vertices then forces its two ends to be both in V or both out, and the lattice is connected, so V is empty. Dually, a product of the plaquettes in a set F is $\prod Z$ over the edges in an odd number of faces of F; a top edge lies in one face only, so no top face is in F, and connectedness through the edges that lie in two faces empties F. A product of stars and plaquettes is a multiple of $\mathbb{1}$ only if its X-part and Z-part are, so the generators are independent and $-\mathbb{1}$ is not generated: $r = n$, and the ground space has dimension $2^0 = 1$.

(ii) Both rough: $n = W(\ell+1) + W(\ell+2)$, with $W(\ell+1)$ stars and $W\ell + 2W$ plaquettes, n generators. Stars: dangling edges again force $V = \emptyset$. Plaquettes: every edge, dangling or

not, lies in exactly two faces, so the product of all plaquettes is $\mathbb{1}$, and by connectedness this is the only relation. So $r = n - 1$, and the dimension is 2. Both smooth: $n = W(\ell + 1) + W\ell$, with $W(\ell + 1)$ stars and $W\ell$ plaquettes, n generators; now every edge has two ends, so the product of all stars is $\mathbb{1}$, the only relation, while the top and bottom edges lie in one face each and the plaquettes are independent. Again $r = n - 1$ and dimension 2. (In both cases the one relation is a product of $\mathbb{1}$'s, so $-\mathbb{1}$ is not generated.)

(iii) $\mathrm{Hom}(1 \oplus e, 1 \oplus m)$ is one-dimensional (only 1 in common), $\mathrm{Hom}(1 \oplus e, 1 \oplus e)$ and $\mathrm{Hom}(1 \oplus m, 1 \oplus m)$ are two-dimensional: $1, 2, 2$, the ground-state counts. The general rule, the ground-state degeneracy is $\dim \mathrm{Hom}(L_{\mathrm{top}}, L_{\mathrm{bottom}})$, is that of Lan, Wang and Wen (Phys. Rev. Lett. 114 (2015) 076402, arXiv:1408.6514), cited: an anyon that condenses on both boundaries can be created at one and absorbed at the other, without a trace. With both ends rough, a Z-string from the bottom to the top boundary violates nothing ((i) of Exercise 1, at both ends) and commutes with every term; so does an X-loop around the cylinder on a dual cycle. They share one edge, so they anticommute: on the two ground states they act as Z and X of a qubit, the e tunnelling between the boundaries and the m circling the cylinder.